The documentation and process conversion measures necessary to comply with this document shall be completed by 16 September 2023.

INCH-POUND

MIL-PRF-19500/391R
w/AMENDMENT 5
16 June 2023
SUPERSEDING
MIL-PRF-19500/391R
w/AMENDMENT 4
9 December 2022

PERFORMANCE SPECIFICATION SHEET

TRANSISTOR, NPN, SILICON, LOW-POWER, DEVICE TYPES 2N3019, 2N3057A, AND 2N3700, ENCAPSULATED (THROUGH-HOLE AND SURFACE MOUNT) AND UNENCAPSULATED, RADIATION HARDNESS ASSURANCE, QUALITY LEVELS JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN, silicon, low-power transistors. Four levels of product assurance (JAN, JANTX, JANTXV and JANS) are provided for each device type as specified in MIL-PRF-19500. Two levels of product assurance (JANHC and JANKC) are provided for each unencapsulated device type. Provisions for radiation hardness assurance (RHA) to eleven radiation levels are provided for quality levels JANTXV, JANS, JANHC, and JANKC. RHA level designators "E", "K", "U", "M", "D", "P", "L", "R", "F", "G", and "H" are appended to the device prefix to identify devices, which have passed RHA requirements.

1.2 Physical dimensions.

1.2.1 Package outlines. The device packages for the encapsulated device types are as follows: TO-205AA and TO-205AD (formerly TO-5 and TO-39) in accordance with figure 1, TO-206AB (formerlyTO-46) in accordance with figure 2, three terminal round metal can TO-206AA (formerly TO-18) in accordance with figure 3, and four terminal SMD package UB and UBC in accordance with figure 4.

1.2.2 Un-encapsulated die. The dimensions and topography for JANHC and JANKC unencapsulated die are as follows: The A version die (JANHCA and JANKCA) is in accordance with figure 5, the B version die (JANHCB and JANKCB) is in accordance with figure 6, the C version die (JANHCC and JANKCC) is in accordance with figure 7, and the D version die (JANHCD and JANKCD) is in accordance with figure 8.

1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

Types	I_C	V_{CBO}	V_{EBO}	V_{CEO}	T_J and T_{STG}
	A dc	V dc	V dc	V dc	°C
All devices	1	140	7	80	-65 to +200

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1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C - Continued.

Types	P_T T_A = +25°C (1) (2)	P_T T_C = +25°C (1) (2)	P_T $T_{SP(IS)}$ = +25°C (1) (2)	$R_{\theta JA}$ (2) (3) (4)	$R_{\theta JC}$ (2) (3)	$R_{\theta JSP(IS)}$ (2) (3)
	W	W	W	°C/W	°C/W	°C/W
2N3019	0.800	5	N/A	195	30	N/A
2N3019S	0.800	5	N/A	195	30	N/A
2N3057A	0.500	1.8	N/A	325	80	N/A
2N3700	0.500	1	N/A	325	150	N/A
2N3700UB, UBC	0.500	N/A	1.5	325	N/A	90

(1) For derating, see, figures 9, 10, 11, 12, 13, and 14.
(2) See 3.3.
(3) For thermal curves, see figures 15, 16, 17, 18, 19, 20, and 21.
(4) For non-thermal conductive PCB or unknown PCB surface mount conditions in free air, substitute figures 9 and 13 for the UB and UBC package and use $R_{\theta JA}$.

1.4 Primary electrical characteristics.

Limit	h_{FE1} V_{CE} = 10 V dc I_C = 150 mA dc	h_{FE2} V_{CE} = 10 V dc I_C = 0.1 mA dc	h_{FE3} (1) V_{CE} = 10 V dc I_C = 10 mA dc	h_{FE4} (1) V_{CE} = 10 V dc I_C = 500 mA dc	h_{FE5} (1) V_{CE} = 10 V dc I_C = 1 A dc
Min	100	50	90	50	15
Max	300	300		300	

Limit	$\lvert h_{fe} \rvert$ f = 20 MHz V_{CE} = 10 V dc I_C = 50 mA dc	C_{obo} 100 kHz ≤ f ≤ 1 MHz V_{CB} = 10 V dc I_E = 0	$V_{CE(sat)1}$ (1) I_C = 150 mA dc I_B = 15 mA dc	$V_{CE(sat)2}$ (1) I_C = 500 mA dc I_B = 50 mA dc	$V_{BE(sat)}$ (1) I_C = 150 mA dc I_B = 15 mA dc
		pF	V dc	V dc	V dc
Min	5				
Max	20	12	0.2	0.5	1.1

(1) Pulsed, see 4.5.1.

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN brand and quality level designators.

1.5.1.1 Encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", "JANTXV" and "JANS".

1.5.1.2 Unencapsulated die. The quality level designators for unencapsulated die that are applicable for this specification sheet from the lowest to the highest level are as follows: "JANHC" and "JANKC".

1.5.2 RHA designator. The RHA levels applicable for this specification sheet from lowest to highest for quality levels JANS and JANKC are as follows: "E", "K", "U", "M", "D", "P", "L", "R", "F", "G", and "H". The RHA levels applicable for this specification sheet from lowest to highest for JANTXV quality levels are as follows: "R" and "F".

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1.5.3 Device type. The designation system for the device types of transistors covered by this specification sheet are as follows.

1.5.3.1 First number and first letter symbols. The transistors of this specification sheet use the first number and letter symbols "2N".

1.5.3.2 Second number symbols. The second number symbols for the transistors covered by this specification sheet are as follows: "3019", "3057", and "3700".

1.5.4 Suffix symbols. The following suffix letters are incorporated in the PIN as applicable.

1.5.4.1 Modified version designator. Device type 2N3057 is available only with an "A" suffix symbol. The "A" suffix symbol indicates an electrical parameter modified version of the device versus the non-A suffix device.

1.5.4.2 Package designators. The suffix symbols (or lack thereof) that designate the package outline for the devices covered by this specification sheet are as follows:

	For device type 2N3019 only, a blank package designator identifies the package as a TO-205AA (see figure 1) with a dimension LL range from 1.50 minimum to 1.75 maximum.
S	For device type 2N3019 only, a "S" package designator identifies the package as a TO-205AD (see figure 1) with a dimension LL range from .500 minimum to .750 maximum.
	For device type 2N3057A only, a blank package designator identifies the package as a TO-206AB (see figure 2).
	For device type 2N3700 only, a blank package designator identifies the package as a TO-206AA (see figure 3).
UB	For device type 2N3700 only, a "UB" package designator identifies that the package is a 4-terminal metal lid (used as a shield and connected to fourth pad) SMD package (see figure 4).
UBC	For device type 2N3700 only, a "UBC" package designator identifies that the package is a 4-terminal ceramic lid SMD package (Fourth pad connected to the lid braze ring) (see figure 4).

1.5.5 Lead finish. The lead finishes applicable to this specification sheet are listed on QML-19500.

1.5.6 Die identifiers for unencapsulated devices. The manufacturer die identifiers that are applicable for this specification sheet are "A" (see figure 5), "B" (see figure 6), "C" (see figure 7), and "D" (see figure 8).

1.6 Radiation features. The following radiation features are applicable for RHA devices supplied to this specification sheet.

1.6.1 Maximum total ionizing dose (TID). The maximum TID that RHA devices were tested to in accordance with condition A (dose rate = 50 to 300 rad(Si)/s) of method 1019 of MIL-STD-750 are as follows:

For device type 2N3019, 2N3019S, 2N3057A, 2N3700, 2N3700UB, AND 2N3700UBC:..... 300 krads(Si) 1/

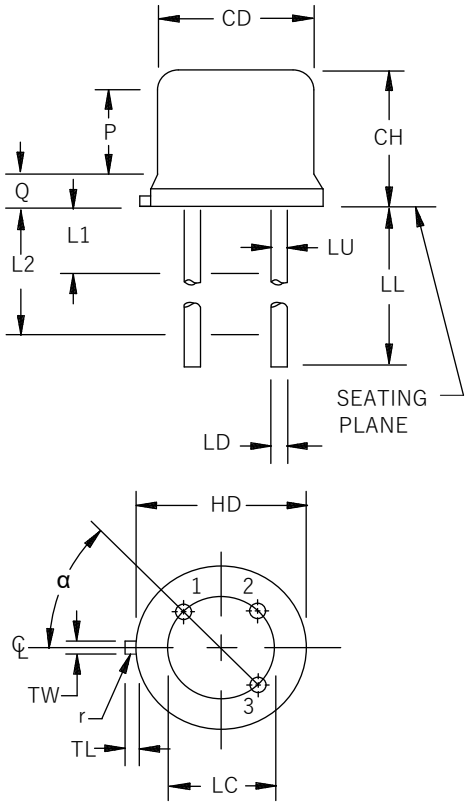
* 1.6.2 Maximum total ionizing dose (TID). The maximum TID that RHA devices were tested to in accordance with condition D (dose rate ≤ 10 mrad(Si)/s)of method 1019 of MIL-STD-750 are as follows:

For device type 2N3019, 2N3019S, 2N3057A, 2N3700, 2N3700UB AND 2N3700UBC:..... 100 krads(Si) 2/

1/ The manufacturers supplying these device types have performed characterization testing in accordance with condition A (dose rate = 50 to 300 rad(Si)/s) of method 1019 of MIL-STD-750. The radiation end point limits are guaranteed only to a maximum TID level of 300 krads(Si).
2/ The manufacturers supplying these device types have performed characterization testing in accordance with condition D (dose rate ≤ 10 mrad(Si)/s) of method 1019 of MIL-STD-750. The radiation end point limits are guaranteed only to a maximum TID level of 100 krads(Si).

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Symbol	Dimensions				Notes
	in	in	mm	mm	
	Min	Max	Min	Max	
CD	.305	.335	7.75	8.51	4
CH	.240	.260	6.10	6.60	
HD	.335	.370	8.51	9.40	4
LC	.200 TP		5.08 TP		5
LD	.016	.021	0.41	0.53	6, 7
LL	1.50	1.750	38.10	40.45	6, 7, 8
LU	.016	.019	0.41	0.48	6, 7
L1		.050		1.27	6, 7
L2	.250		6.35		6, 7
P	.100		2.54		
Q		.050		1.27	4
TL	.029	.045	0.74	1.14	9
TW	.028	.034	0.71	0.86	10
r		.010		0.25	11
α	45 TP		45 TP		5



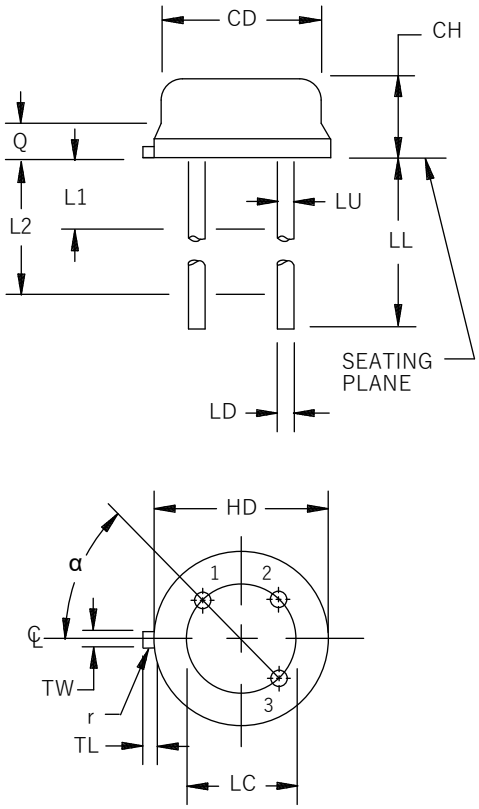
NOTES:

1. Dimension are in inches. Millimeters are given for general information only.
2. Terminal 1 = emitter, terminal 2 = base, terminal 3 = collector.
3. The collector shall be internally connected to the case.
4. Body contour optional within zone defined by dimensions CD, HD, and Q.
5. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods.
6. Dimension LU applies between dimensions L1 and L2. Dimension LD applies between dimensions L2 and LL minimum. Diameter is uncontrolled in dimension L1 and beyond dimension LL minimum.
7. All three leads.
8. For "S" suffix devices, dimension LL is .500 (12.70 mm) minimum, .750 (19.05 mm) maximum.
9. Dimension TL measured from maximum HD.
10. Beyond r (radius) maximum, dimension TW shall be held for a minimum length of .011 inch (0.28 mm).
11. Dimension r (radius) applies to both inside corners of tab.

FIGURE 1. Physical dimensions for TO-205AA and AD packages (formerly TO-5) (device type 2N3019 only).

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Symbol	Dimensions				Notes
	in	in	mm	mm	
	Min	Max	Min	Max	
CD	.178	.195	4.52	4.95	
CH	.065	.085	1.65	2.16	
HD	.209	.230	5.31	5.84	
LC	.100 TP		2.54 TP		4
LD	.016	.021	0.41	0.53	5
LL	.500	1.750	12.70	44.45	5, 6
LU	.016	.019	0.41	0.48	5, 6
L1		.050		1.27	5, 6
L2	.250		6.35		5, 6
Q		.040		1.02	
TL	.028	.048	0.71	1.22	7
TW	.036	.046	0.91	1.17	8, 9
r		.010		0.25	9, 10
α	45 TP		45 TP		



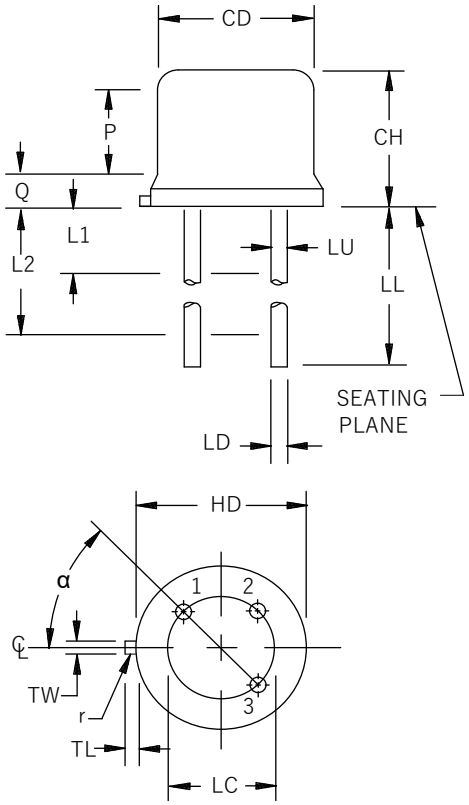
NOTES:

1. Dimensions are in inches. Millimeters are given for general information only.
2. Terminal 1 is emitter, terminal 2 is base, and terminal 3 is collector.
3. Terminal 3 (collector) is electrically connected to case.
4. Leads at gauge plane .054 inch (1.37 mm) +.001 inch (0.03 mm) -.000 inch (0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of TP relative to tab. Device may be measured by direct methods or by gauge.
5. All three leads.
6. Dimension LU applies between L1 and L2. Dimension LD applies between L2 and LL minimum.
7. Dimension TL is measured from dimension HD maximum.
8. Details of outline in this zone are optional.
9. Beyond r (radius) maximum, dimension TW shall be held for a minimum length of .011 inch (0.28 mm).
10. Dimension r (radius) applied to both inside corners of tab.

FIGURE 2. Physical dimensions for TO-206AB package (formerly TO-46) (device type 2N3057A only).

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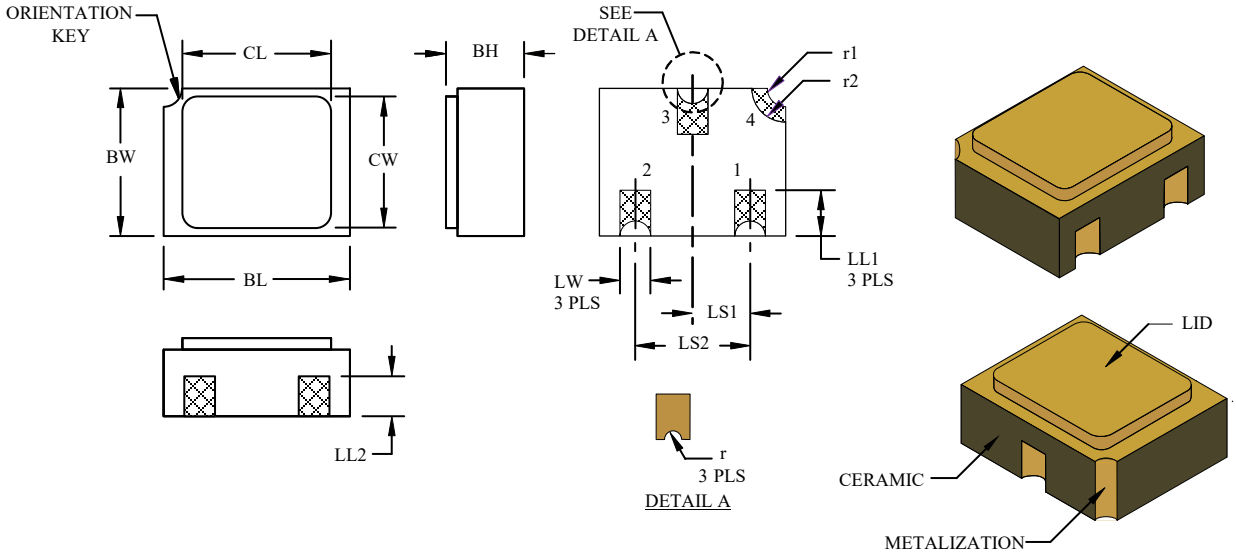
Symbol	Dimensions				Note
	in	in	mm	mm	
	Min	Max	Min	Max	
CD	.178	.195	4.52	4.95	4
CH	.170	.210	4.32	5.33	
HD	.209	.230	5.31	5.84	4
LC	.100 TP		2.54 TP		5
LD	.016	.021	0.41	0.53	6, 7
LL	.500	.750	12.70	19.05	6, 7
LU	.016	.019	0.41	0.48	6, 7
L1		.050		1.27	6, 7
L2	.250		6.35		6, 7
P	.100		2.54		
Q		.030		0.76	4
TL	.028	.048	0.71	1.22	8
TW	.036	.046	0.91	1.17	9
r		.010		0.25	10
α	45 TP		45 TP		5



NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Terminal 1 = emitter, terminal 2 = base, terminal 3 = collector.
3. The collector shall be internally connected to the case.
4. Body contour optional within zone defined by dimensions CD, HD, and Q.
5. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods or by gauge
6. Dimension LU applies between dimensions L1 and L2. Dimension LD applies between dimensions L2 and LL minimum. Diameter is uncontrolled in dimension L1 and beyond dimension LL minimum.
7. All three leads.
8. Dimension TL measured from maximum HD.
9. Beyond r (radius) maximum, dimension TW shall be held for a minimum length of .011 inch (0.28 mm).
10. Dimension r (radius) applies to both inside corners of tab.

FIGURE 3. Physical dimensions for TO-206AA package (formerly TO-18) (device type 2N3700 only).

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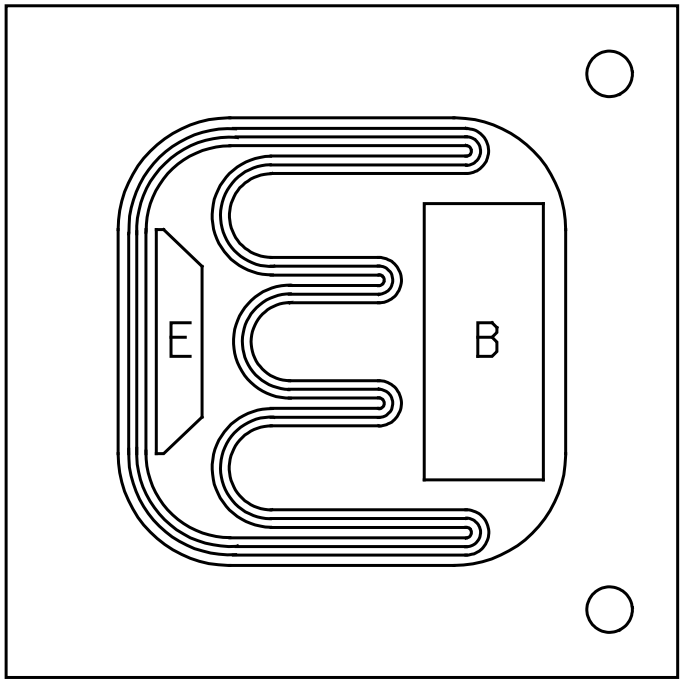


Symbol	Dimensions				Note
	in	in	mm	mm	
	Min	Max	Min	Max	
BH	.040	.056	1.02	1.42	
BL	.115	.128	2.92	3.25	
BW	.085	.108	2.16	2.74	
CL		.128		3.25	
CW		.108		2.74	
LL1	.022	.038	0.56	0.96	
LL2	.017	.035	0.43	0.89	
LS1	.035	.040	0.89	1.02	
LS2	.071	.079	1.81	2.01	
LW	.016	.024	0.41	0.61	
r		.008		.203	
r1		.012		.305	
r2		.022		.559	

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Hatched areas on package denote metallized areas.
3. Pad 1 = Base, Pad 2 = Emitter, Pad 3 = Collector, Pad 4 = Shielding connected to the lid.
4. UBC (ceramic lid) only: Pad 1 = Base, Pad 2 = Emitter, Pad 3 = Collector, Pad 4 = Connected to the lid braze ring.

FIGURE 4. Physical dimensions for 4 terminal SMD packages (device type 2N3700UB and 2N3700UBC only).

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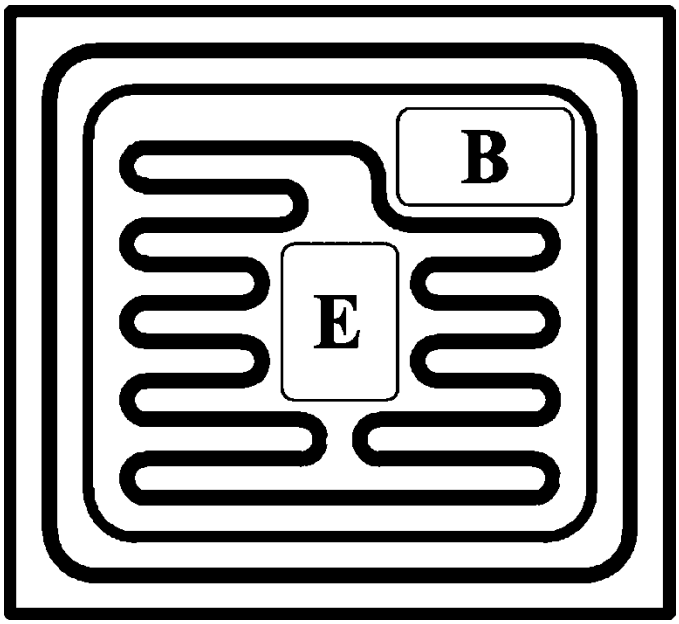


(A-version)

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. The physical characteristics of the die are:
   a. Die size: .030 x .030 inch (0.762 x 0.762 mm).
   b. Die thickness: .008 ±.0016 inch (0.2032 mm ±0.04064 mm).
   c. Base pad: .004 x .010 inch (0.1016 mm x 0.254 mm).
   d. Emitter pad: .0023 x .007 inch (0.05842 mm x 0.1778 mm).
   e. Collector pad: Back side.
3. The metallization characteristics of the die are:
   a. Top metal: Aluminum, 12,000 Å minimum; 14,500 Å nominal.
   b. Backside metal: Gold, 4,000 ±1,000 Å
4. Glassivation: SiO_2, 6,000 ±1,000 Å.

FIGURE 5. JANHCA and JANKCA die (2N3700 and 2N3019) dimensions.

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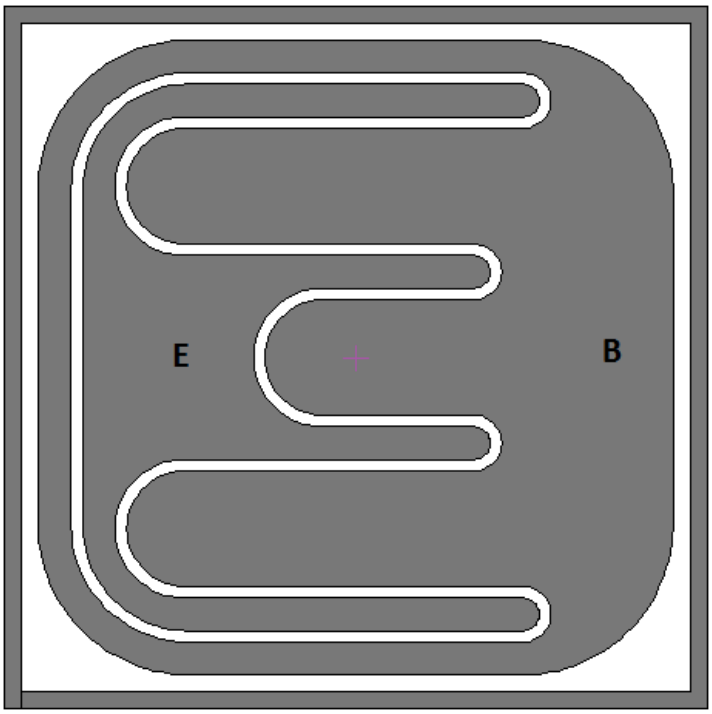
(B-version)

NOTES:

1. Dimensions are in inches. Millimeters are given for general information only.
2. The physical characteristics of the die are:
   a. Die size: .024 x .026 inch ±.002 inch (0.610 X 0.660 ±0.051 millimeter).
   b. Die thickness: .010 inch ±.0015 inch (0.254 ±0.0381 millimeter).
   c. Base pad: B = .004 inch x .006 inch (0.102 X 0.152 millimeter).
   d. Emitter pad: E = .004 inch x .0055 inch (0.102 X 0.140 millimeter).
   e. Collector pad: Backside.
3. The metallization characteristics of the die are:
   a. Top metal: Aluminum 15,000Å minimum, 18,000Å nominal.
   b. Backside metal: Gold: 3,500Å minimum, 5,000Å nominal.

FIGURE 6. JANHCB and JANKCB die (2N3700) dimensions.

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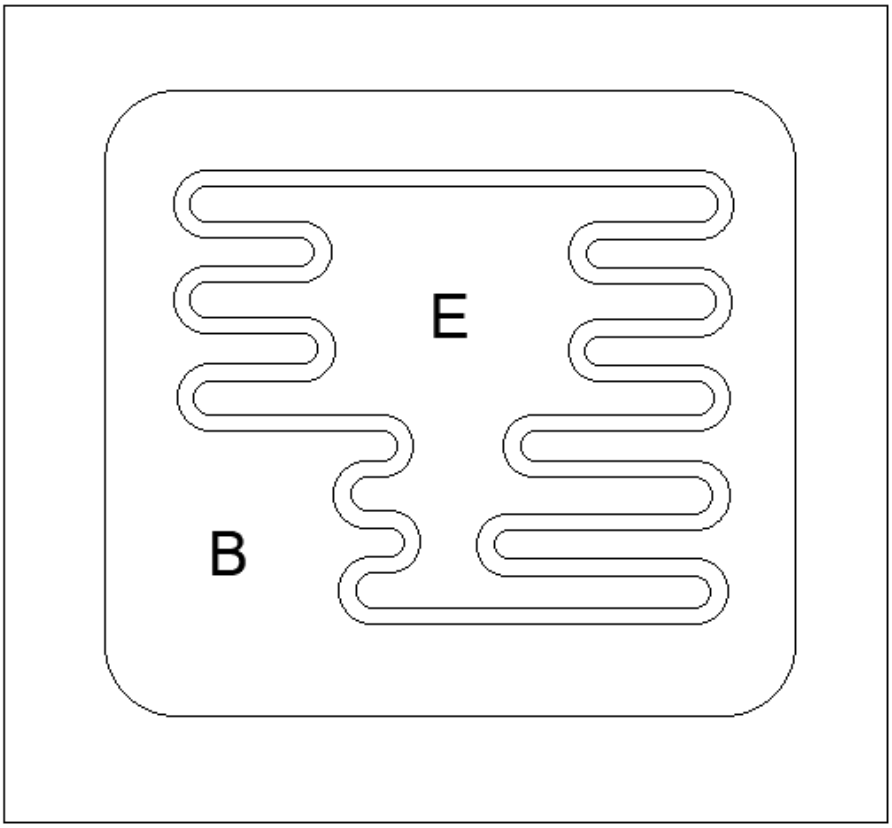
(C-version)

NOTES:

1. Dimensions are in inches. Millimeters are given for general information only.
2. The physical characteristics of the die are:
   a. Die size: .023 x .023 inch ±.002 inch (0.584 X 0.584 ±0.051 millimeter).
   b. Die thickness: .010 inch ±.002 inch (0.254 ±0.508 millimeter).
   c. Base pad: B = .004 inch x .010 inch (0.102 X 0.254 millimeter).
   d. Emitter pad: E = .0039 inch x .0039 inch (0. 992 X 0.099 millimeter).
   e. Collector pad: Backside.
3. The metallization characteristics of the die are:
   a. Top metal: Aluminum 16,000Å minimum, 20,000Å nominal.
   b. Backside metal: Gold: 5,000Å ±500Å.

FIGURE 7. JANHCC and JANKCC die (2N3700) dimensions.

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(D-version)

NOTES:

1. Dimensions are in inches. Millimeters are given for general information only.
2. The physical characteristics of the die are:
   a. Die size: .023 x .023 inch [± .002 inch] (0.584 mm x 0.584 mm [± 0.058mm])
   b. Die thickness: .008 inch [± .0016 Inch] (0.203 mm [± 0.0406mm])
   c. Base pad: .004 x .004 inch (0.109 mm x 0.109 mm)
   d. Emitter pad: .004 x .004 inch (0.109 mm x 0.109 mm
   e. Collector: Backside.
3. The metallization characteristics of the die are:
   a. Top metal: Aluminum, 14,000 Å Minimum, 20,000 Å Nominal
   b. Back metal: Gold, 8,400 Å Minimum, 12,000 Å Nominal

FIGURE 8. JANHCD and JANKCD die (2N3700 and2N3019) dimensions.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML-19500) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

PCB ..... Printed circuit board.
$R_{\theta JA}$ ..... Thermal resistance junction to ambient.
$R_{\theta JC}$ ..... Thermal resistance junction to case.
$R_{\theta JSP(IS)}$..... Thermal resistance junction to solder pads (infinite sink mount to PCB).
$T_{SP(IS)}$..... Temperature of solder pads (infinite sink mount to PCB).
UB, UBC ..... Surface mount case outlines (see figure 4).

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figures 1, 2, 3, 4, 5, 6, 7, and 8 herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Pin-out. The pin-out of the device shall be as shown on figures 1, 2, 3, and 4 as applicable.

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3.5 Marking.

3.5.1 Marking of all devices except "UB" and "UBC" suffix devices. Marking shall be in accordance with MIL-PRF-19500.

3.5.2 Marking of "UB" and "UBC" suffix devices. Marking on the UB and UBC packages shall consist of an abbreviated PIN, the date code, and the manufacturers symbol or logo. The prefixes JAN, JANTX, JANTXV, and JANS can be abbreviated as J, JX, JV, and JS respectively. The "2N" prefix can also be omitted. The RHA designator (see 1.5.2) shall immediately precede (or replace) the device "2N" identifier (depending upon degree of abbreviation required).

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.8 RHA. Radiation hardness assurance requirements and test levels shall be as defined in MIL-PRF-19500.

3.9 Workmanship. Transistors shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I, II, and III).

d. Element evaluation (see 4.6).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 JANHC and JANKC qualification. JANHC and JANKC qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.2 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.2.3 Radiation hardened devices. See 4.4.4 herein and MIL-PRF-19500.

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4.3 Screening of encapsulated devices (quality levels JANTX, JANTXV, and JANS only). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen	Measurements for JANS	Measurements for JANTX and JANTXV
(1) 3c	See 4.3.2.	See 4.3.2.
9	I_{CES1} and h_{FE1}.	Not applicable.
10	48 hours minimum.	48 hours minimum.
11	I_{CES1}; h_{FE1}; ΔI_{CES1} = 100 percent of initial value or 5 nA dc, whichever is greater; Δh_{FE1} = ±15 percent.	I_{CES1} and h_{FE1}.
12	See 4.3.1.	See 4.3.1.
13	Subgroups 2 and 3 of table I herein; ΔI_{CES1} = 100 percent of initial value or 5 nA dc, whichever is greater; Δh_{FE1} = ±15 percent.	Subgroup 2 of table I herein; ΔI_{CES1} = 100 percent of initial value or 5 nA dc, whichever is greater; Δh_{FE1} = ±15 percent.

(1) Shall be performed anytime after temperature cycling, screen 3a; JANTX and JANTXV levels do not need to be repeated in screening requirements.

4.3.1 Power burn-in conditions. Power burn-in conditions shall be as follows: V_{CB} = 10 - 30 V dc. Power shall be applied to achieve T_J = +135°C minimum using a minimum P_D = 75 percent of P_T maximum, T_A ambient rated as defined in 1.3. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, T_J, and mounting conditions) may be used for JANTX and JANTXV quality levels. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval. Use method 3100 of MIL-STD-750 to measure T_J.

4.3.2 Thermal impedance measurements. The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{MD} (and V_C where appropriate). See table III, subgroup 4 herein.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. If alternate screening is being performed in accordance with MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of group A1 and A2 inspection only (table E-VIB, group B, subgroup 1 is not required to be performed again if group B has already been satisfied in accordance with 4.4.2).

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with 4.4.2.1 for quality level JANS and 4.4.2.2 for quality levels JAN, JANTX, and JANTXV.

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4.4.2.1 Group B inspection (JANS). Group B inspection for quality level JANS shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIA of MIL-PRF-19500 and herein.

Subgroup	Method	Condition
B4	1037	V_{CB} = 10 V dc, 2,000 cycles, adjust device current, or power, to achieve a minimum ΔT_J of +100°C.
B5	1027	NOTE: If a failure occurs, resubmission shall be at the test conditions of the original sample. V_{CB} = 10 V dc, P_D ≥ 100 percent of maximum rated P_T (see 1.3). Option 1: 96 hours minimum, sample size in accordance with MIL-PRF-19500, table E-VIA, adjust T_A or P_D to achieve T_J = +275°C minimum. Option 2: 216 hours minimum, sample size = 45, c = 0; adjust T_A or P_D to achieve T_J = +225°C minimum.
B5	2037	Test condition D.
B6		Not applicable.

4.4.2.2 Group B inspection (JAN, JANTX, and JANTXV). Group B inspection for quality levels JAN, JANTX, and JANTXV shall be conducted in accordance with the conditions specified in table E-VIC (small die flow) of MIL-PRF-19500 and herein. All catastrophic failures during conformance inspection shall be analyzed to the extent possible to identify root cause and corrective action. Whenever a failure is identified as wafer lot or wafer processing related, the entire wafer lot and related devices assembled from the wafer lot shall be rejected unless an appropriate determined corrective action to eliminate the failure mode has been implemented and the devices from the wafer lot are screened to eliminate the failure mode.

Step	Method	Condition
1	1026	Steady-state life: 1,000 hours minimum, V_{CB} = 10 V dc, power shall be applied to achieve T_J = +150°C minimum using a minimum of P_D = 75 percent of maximum rated P_T as defined in 1.3. n = 45 devices, c = 0. The sample size may be increased and the test time decreased so long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours.
2	1048	Blocking life: T_A = +150°C, V_{CB} = 80 percent rated voltage, without exceeding max rated V_{CEO}. 48 hours minimum. n = 45 devices, c = 0.
3	1032	High-temperature life (non-operating), t = 340 hours, T_A = +200°C. n = 22, c = 0.

4.4.2.2.1 Sample selection. Samples selected for small die flow group B inspection shall be in accordance with all of the following requirements:

a. Samples shall be selected from an inspection lot that has been submitted to and passed table I, subgroup 2, conformance inspection.

b. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (step 1) may be tested prior to the application of final lead finish.

c. Separate samples may be used for each step.

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4.4.2.2.2 Failures. In the event of a lot failure, the resubmission requirements of MIL-PRF-19500 shall apply. In addition, all catastrophic failures during conformance inspection shall be analyzed to the extent possible to identify root cause and corrective action. Whenever a failure is identified as wafer lot and/or wafer processing related, the entire wafer lot and related devices assembled from the wafer lot shall be rejected unless an appropriate determined corrective action to eliminate the failure~~s~~ mode has been implemented and the devices from the wafer lot are screened to eliminate the failure mode.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and 4.4.3.1 for quality level JANS and 4.4.3.2 for quality levels JAN, JANTX, and JANTXV.

4.4.3.1 Quality level JANS.

Subgroup	Method	Condition
C2	2036	Test condition E; not applicable for UB and UBC devices.
C5	3131	See 1.3. $R_{\theta JC}$ is applicable for all devices except UB and UBC. $R_{\theta JA}$ is applicable for UB and UBC devices.
C6	1026	1,000 hours, V_{CB} = 10 V dc, power and ambient temperature shall be applied to the device to achieve T_J = +150°C minimum, and minimum power dissipation of 75 percent of max rated P_T (see 1.3 herein); n = 45, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours.

4.4.3.2 Quality levels JAN, JANTX, and JANTXV.

Subgroup	Method	Condition
C2	2036	Test condition E; not applicable for UB and UBC devices.
C5	3131	See 1.3, $R_{\theta JC}$ is applicable for all devices except UB and UBC. $R_{\theta JA}$ is applicable for UB and UBC devices.
C6		Not applicable.

4.4.3.3 Sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes table I tests herein for conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for C6 life test may be tested prior to the application of final lead finish. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.4.4 Group D inspection. Conformance inspection for radiation hardness assured JANS and JANTXV types shall include the group D tests specified in table II herein. These tests shall be performed as required in accordance with MIL-PRF-19500 and method 1019 of MIL-STD-750 for total ionizing dose, or method 1017 of MIL-STD-750 for neutron fluence as applicable (see 6.2.e herein), except group D, subgroup 2 may be performed separate from other subgroups. Alternate package options may also be substituted for the testing provided there is no adverse effect to the fluence profile.

4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 as specified in table III herein.

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4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Collector-base time constant. This parameter may be determined by applying an rf signal voltage of 1.0 volt (rms) across the collector-base terminals, and measuring the ac voltage drop (V_{eb}) with a high- impedance rf voltmeter across the emitter-base terminals. With f = 79.8 MHz used for the 1.0 volt signal, the following computation applies:

$$r'_b , C_{c(ps)} = 2 X V_{eb} \text{ (millivolts)}$$

4.5.3 Displacement damage characterization. For RHA devices, each supplier shall perform a displacement damage characterization. The characterization shall demonstrate exposure versus data and does not indicate pass or fail criteria. The exposure shall be conducted in accordance with method 1017 of MIL-STD-750. The following details shall apply:

a. Samples may be taken from any wafer of the qualification lot.

b. As a minimum, testing shall be at 2E+12 n/cm2 plus two additional fluence levels chosen by the manufacturer.

c. If the device degrades less than 5 percent of the specification at the highest fluence level, a single data point may be sufficient.

Alternate package options may be substituted for characterization. The displacement damage characterization data shall be made available from the manufacturer.

4.6 Element evaluation of unencapsulated die. The element evaluation of unencapsulated die shall be in accordance with appendix G of MIL-PRF-19500. For subgroup 4, the burn-in duration for the JANKC level shall follows the JANS requirements and the JANHC shall follows the JANTX requirements.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit	Limit	Unit
	Method	Conditions		Min	Max	
Subgroup 1 2/						
Visual and mechanical examination	2071					
Solderability 3/ 4/	2026	n = 15 leads, c = 0				
Resistance to solvents 3/ 4/ 5/	1022	n = 15 devices, c = 0				
Salt atmosphere (corrosion) 6/	1041	n = 6 devices, c = 0				
Temperature cycling (air to air) 3/ 4/	1051	Test condition C, 25 cycles n = 22 devices, c = 0				
Hermetic seal 4/ Fine and gross leak	1071	n = 22 devices, c = 0				
Electrical measurements 4/		Table I, subgroup 2				
Bond strength 3/ 4/	2037	Precondition T_A = +250°C at t = 24 hrs or T_A = +300°C at t = 2 hrs, n = 11 wires, c = 0				
Decap internal visual (design verification) 4/	2075	n = 4 devices, c = 0				
Subgroup 2.						
Thermal impedance 7/	3131	See 4.3.2	$Z_{\theta JX}$			°C/W
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 140 V dc	I_{CBO1}		10	µA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 7 V dc	I_{EBO1}		10	µA dc
Collector to emitter breakdown voltage	3011	Bias condition D; I_C = 30 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	80		V dc
Collector to emitter cutoff current	3041	Bias condition C; V_{CE} = 90 V dc	I_{CES1}		10	nA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 5 V dc	I_{EBO2}		10	nA dc
Forward current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 150 mA dc; pulsed (see 4.5.1)	h_{FE1}	100	300	

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit	Limit	Unit
	Method	Conditions		Min	Max	
Subgroup 2 – continued.						
Forward current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 0.1 mA dc; pulsed (see 4.5.1)	h_{FE2}	50	300	
Forward current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 10 mA dc; pulsed (see 4.5.1)	h_{FE3}	90		
Forward current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 500 mA dc; pulsed (see 4.5.1)	h_{FE4}	50	300	
Forward current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 1 A dc; pulsed (see 4.5.1)	h_{FE5}	15		
Collector to emitter voltage (saturated)	3071	I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)1}$		0.2	V dc
Collector to emitter voltage (saturated)	3071	I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)2}$		0.5	V dc
Base to emitter voltage (saturated)	3066	Test condition A; I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)}$		1.1	V dc
Subgroup 3						
High-temperature operation		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias condition C; V_{CE} = 90 V dc	I_{CES2}		5	µA dc
Low-temperature operation		T_A = -55°C				
Forward current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 150 mA dc; pulsed (see 4.5.1)	h_{FE6}	40		
Subgroup 4						
Small-signal short- circuit forward-current transfer ratio	3206	V_{CE} = 5 V dc; I_C = 1 mA dc; f = 1 kHz	h_{fe}	80	400	
Magnitude of small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 50 mA dc; f = 20 MHz	$\|h_{fe}\|$	5	20	
Input capacitance (output open circuited)	3240	V_{EB} = 0.5 V dc; I_C = 0; 100 kHz ≤ f ≤ 1 MHz	C_{ibo}		60	pF
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		12	pF

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit	Limit	Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Noise figure	3246	V_{CE} = 10 V dc; I_C = 100 µA dc; R_g = 1 kΩ; power bandwidth = 200 Hz f = 1 kHz	NF		4	dB
Collector to base time constant		V_{CB} = 10 V dc; I_C = 10 mA dc; f = 79.8 MHz (see 4.5.2)	r'_b, C_c		400	ps
Pulse response		See figure 22	$t_{on} + t_{off}$		30	ns
Subgroup 5						
Safe operating area (continuous dc)	3051	T_C = +25°C; t = 10 ms, 1 cycle (See figure 23)				
Test 1		V_{CE} = 10 V dc				
2N3019, 2N3019S		I_C = 500 mA dc				
2N3057A, 2N3700, 2N3700UB, 2N3700UBC		I_C = 180 mA dc				
Test 2		V_{CE} = 40 V dc				
2N3019, 2N3019S		I_C = 130 mA dc				
2N3057A, 2N3700, 2N3700UB, 2N3700UBC		I_C = 50 mA dc				
Test 3		V_{CE} = 80 V dc				
2N3019, 2N3019S		I_C = 60 mA dc				
2N3057A, 2N3700, 2N3700UB, 2N3700UBC		I_C = 30 mA dc				
Electrical measurements		See table I, subgroup 2 herein				
Subgroups 6 and 7						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.
2/ For resubmission of failed test in subgroup 1 of table I, double the sample size of the failed test or sequence of tests. A failure in table I, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for JANS devices.
5/ Not required for laser marked devices.
6/ Required only for laser marked devices. Not required for non-corrosion prone base metals.
7/ For end-point measurements, this test is required for the following subgroups:
Group B, subgroup 3, 4, and 5 (JANS).
Group B, step 1 (JANTX and JANTXV).
Group C, subgroup 2 and 6.

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TABLE II. Group D inspection.

Inspection 1/ 2/ 3/	MIL-STD-750		Symbol	Limit	Limit	Unit
	Method	Conditions		Min	Max	
Subgroup 1 4/						
Neutron irradiation	1017	Neutron exposure V_{CES} = 0 V				
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 140 V dc	I_{CBO1}		20	µA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 7 V dc	I_{EBO1}		20	µA dc
Collector to emitter breakdown voltage	3011	Bias condition D; I_C = 30 mA dc pulsed (see 4.5.1)	$V_{(BR)CEO}$	80		V dc
Collector to emitter cutoff current	3041	Bias condition C; V_{CE} = 90 V dc	I_{CES1}		20	nA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 5 V dc	I_{EBO2}		20	nA dc
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 150 mA dc	[h_{FE1}] 5/	[50]	300	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 0.1 mA dc	[h_{FE2}] 5/	[25]	300	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 10 mA dc	[h_{FE3}] 5/	[45]		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 500 mA dc	[h_{FE4}] 5/	[25]	300	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 1 A dc	[h_{FE5}] 5/	[7.5]		
Collector-emitter saturation voltage	3071	I_C = 150 mA dc; I_B = 15 mA dc (see 4.5.1)	$V_{CE(sat)1}$		.23	V dc
Collector-emitter saturation voltage	3071	I_C = 500 mA dc; I_B = 50 mA dc (see 4.5.1)	$V_{CE(sat)2}$		.58	V dc
Base to emitter voltage (saturated)	3066	Test condition A; I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)}$		1.27	V dc

See footnotes at end of table.

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TABLE II. Group D inspection - Continued.

Inspection 1/ 2/ 3/	MIL-STD-750		Symbol	Limit	Limit	Unit
	Method	Conditions		Min	Max	
Subgroup 2						
Total dose irradiation	1019	Gamma exposure V_{CES} = 64 V Condition A, for High dose rate, Condition D for low dose rate.				
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 140 V dc	I_{CBO1}		20	µA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 7 V dc	I_{EBO1}		20	µA dc
Collector to emitter breakdown voltage	3011	Bias condition D; I_C = 30 mA dc pulsed (see 4.5.1)	$V_{(BR)CEO}$	80		V dc
Collector to emitter cutoff current	3041	Bias condition C; V_{CE} = 90 V dc	I_{CES1}		20	nA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 5 V dc	I_{EBO2}		20	nA dc
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 150 mA dc	[h_{FE1}] 5/	[50]	300	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 0.1 mA dc	[h_{FE2}] 5/	[25]	300	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 10 mA dc	[h_{FE3}] 5/	[45]		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 500 mA dc	[h_{FE4}] 5/	[25]	300	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 1 A dc	[h_{FE5}] 5/	[7.5]		
Collector-emitter saturation voltage	3071	I_C = 150 mA dc; I_B = 15 mA dc; (see 4.5.1)	$V_{CE(sat)1}$		.23	V dc
Collector-emitter saturation voltage	3071	I_C = 500 mA dc; I_B = 50 mA dc; (see 4.5.1)	$V_{CE(sat)2}$		.58	V dc
Base to emitter voltage (saturated)	3066	Test condition A; I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)}$		1.27	V dc

1/ Tests to be performed on all devices receiving radiation exposure.
2/ For sampling plan, see MIL-PRF-19500.
3/ Electrical characteristics apply to the corresponding suffix versions unless otherwise noted.
4/ See 6.2.e herein.
5/ See method 1019 of MIL-STD-750 for how to determine [h_{FE}] by first calculating the delta ($1/h_{FE}$) from the pre- and post-radiation h_{FE}. Notice the [h_{FE}] is not the same as h_{FE} and cannot be measured directly. The [h_{FE}] value can never exceed the pre-radiation minimum h_{FE} that it is based upon.

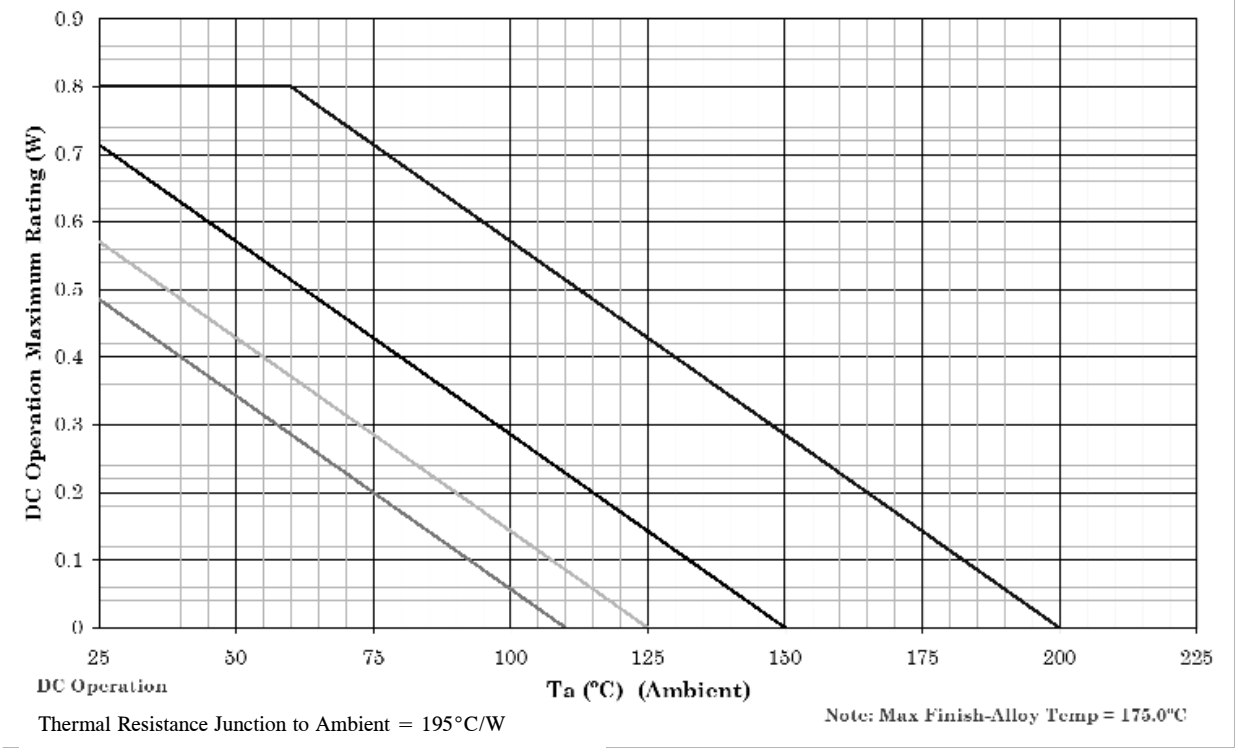
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TABLE III. Group E inspection (all quality levels) - for qualification and re-qualification only.

Inspection	MIL-STD-750		Sample plan
	Method	Conditions	
Subgroup 1			45 devices, c = 0
Temperature cycling (air to air)	1051	Test condition C, 500 cycles.	
Hermetic seal	1071	Fine and gross leak.	
Electrical measurements		See table I, subgroup 2 herein.	
Subgroup 2			45 devices, c = 0
Intermittent life	1037	V_{CB} = 10 V dc, 6,000 cycles.	
Electrical measurements		See table I, subgroup 2 herein.	
Subgroup 4			15 devices, c = 0
Thermal resistance	3131	$R_{\theta JSP(IS)}$ can be calculated but shall be measured once in the same package with a similar die size to confirm calculations (may apply to multiple specification sheets). $R_{\theta JSP(AM)}$ need be calculated only.	
Thermal impedance curves		See table E-IX of MIL-PRF-19500, subgroup 4.	Sample size N/A
Subgroup 5			
Not applicable			
Subgroup 6			
ESD	1020		
Subgroup 8			45 devices, c = 0
Reverse stability	1033	Condition B.	
Subgroup 12			
Neutron irradiation	1017	See 4.5.3.	

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TEMPERATURE-POWER DERATING CURVE
T_A = 25°C; DEVICE TYPE 2N3019



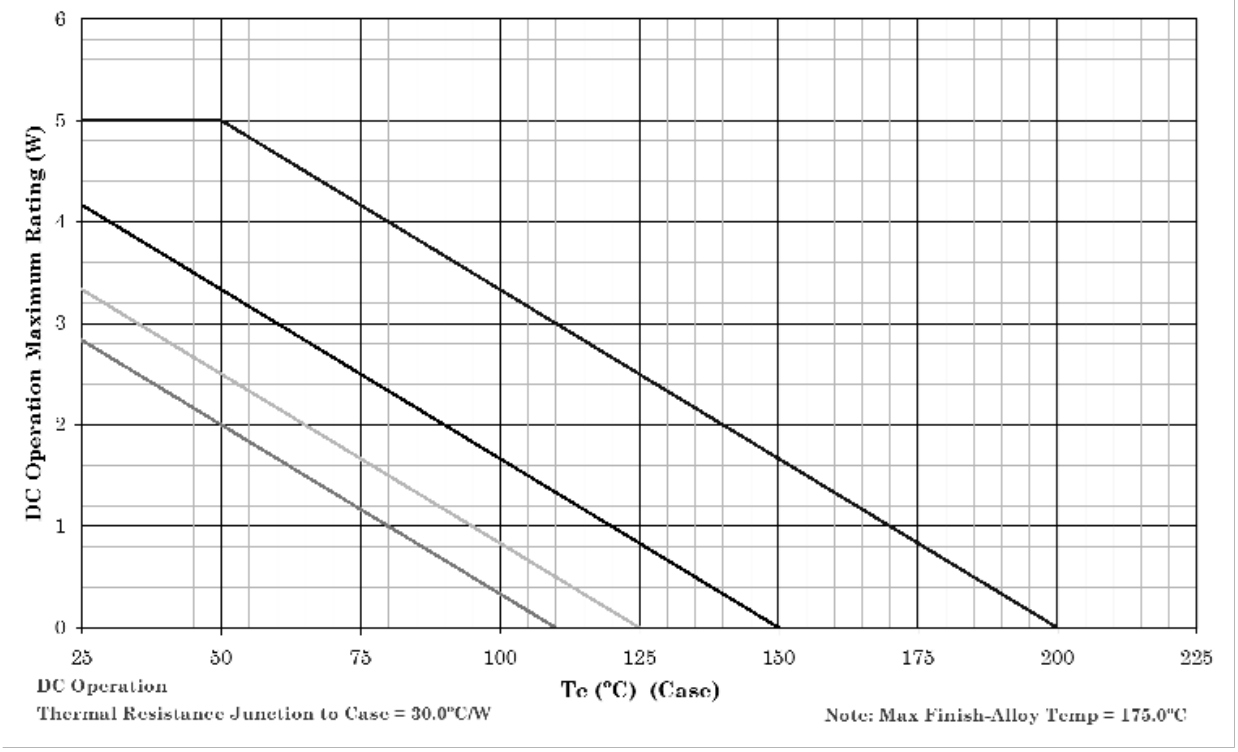
NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperatures and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at T_J ≤ +150°C, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at T_J ≤ +125°C, and +110°C to show power rating where most users want to limit T_J in their application.

FIGURE 9. Derating for 2N3019 ($R_{\theta JA}$) PCB (TO-5 and TO-39).

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Check the source to verify that this is the current version before use.

TEMPERATURE-POWER DERATING CURVE
T_C = 25°C; DEVICE TYPE 2N3019



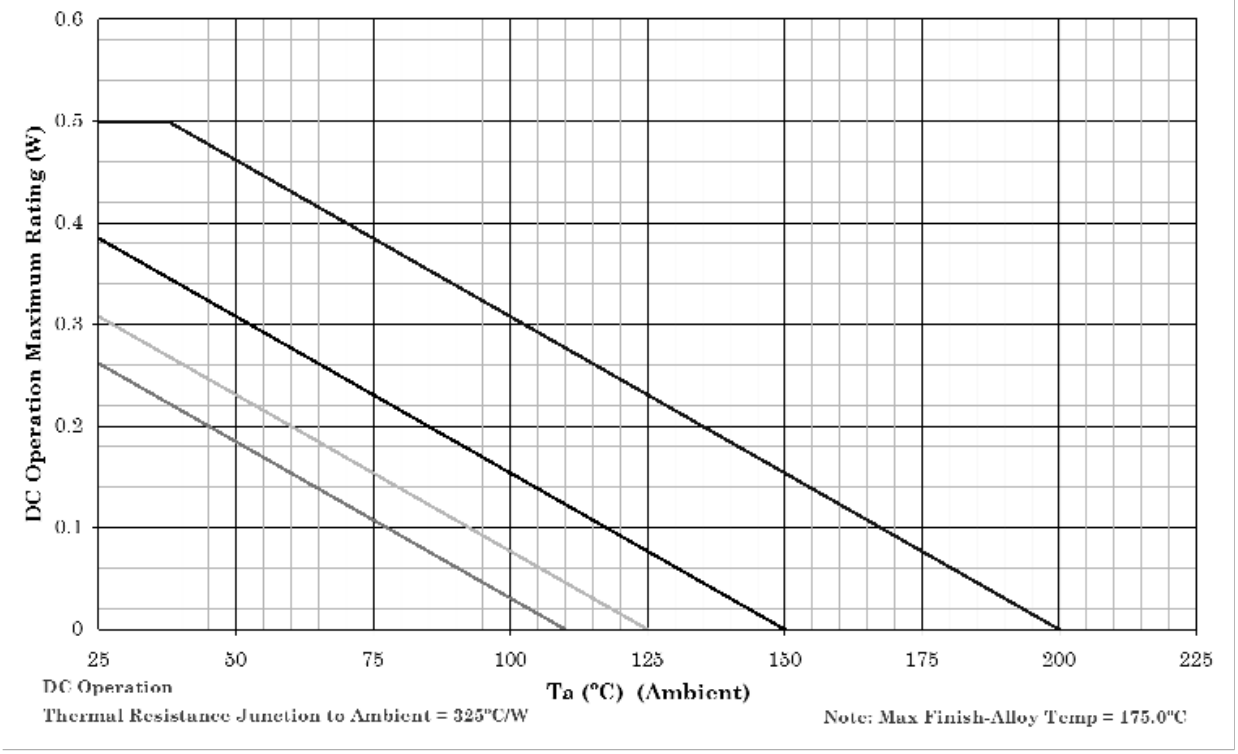
NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperatures and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at T_J ≤ +150°C, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at T_J ≤ +125°C, and +110°C to show power rating where most users want to limit T_J in their application.

FIGURE 10. Derating for 2N3019 ($R_{\theta JC}$), base case mounted (TO-5 and TO-39).

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TEMPERATURE-POWER DERATING CURVE
T_A = 25°C; DEVICE TYPES 2N3057A 2N3700UB AND 2N3700UBC



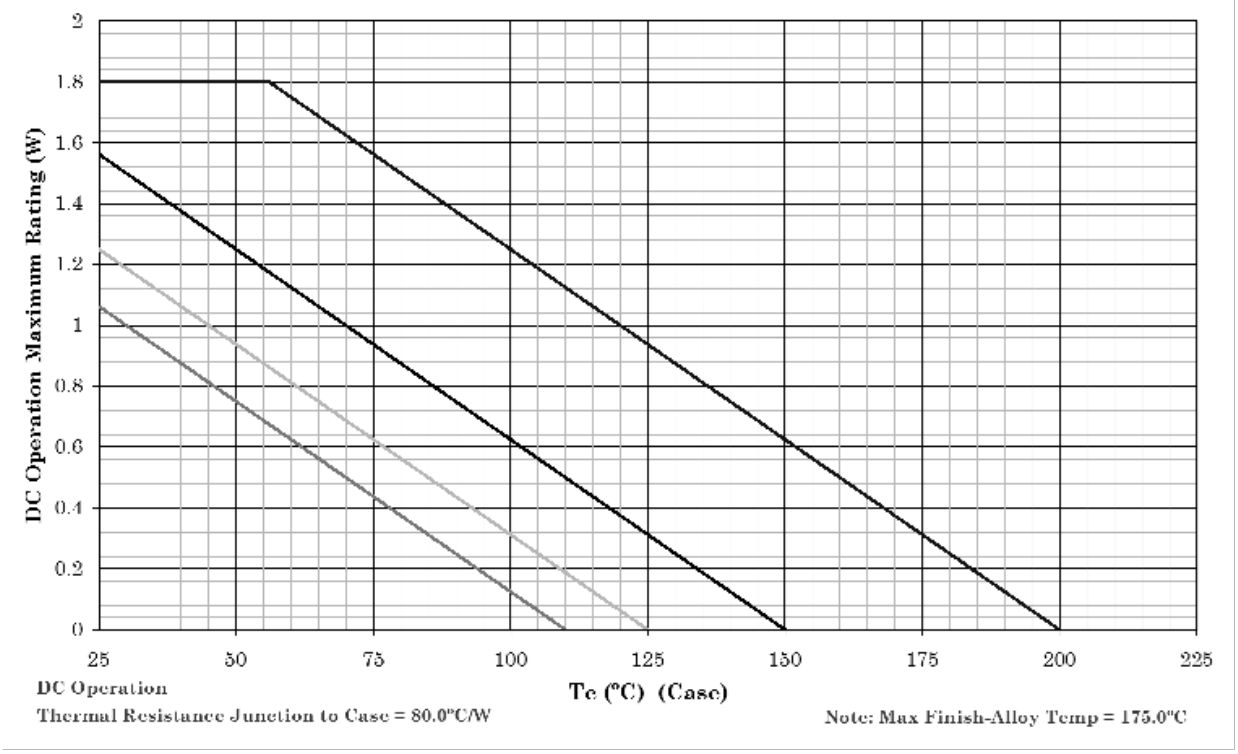
NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperatures and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at T_J ≤ +150°C, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at T_J ≤ +125°C, and +110°C to show power rating where most users want to limit T_J in their application.

FIGURE 11. Derating for 2N3057A, 2N3700, 2N3700UB and 2N3700UBC ($R_{\theta JA}$) leads .125 inch (3.175 mm).

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TEMPERATURE-POWER DERATING CURVE
T_C = 25°C; DEVICE TYPE 2N3057A



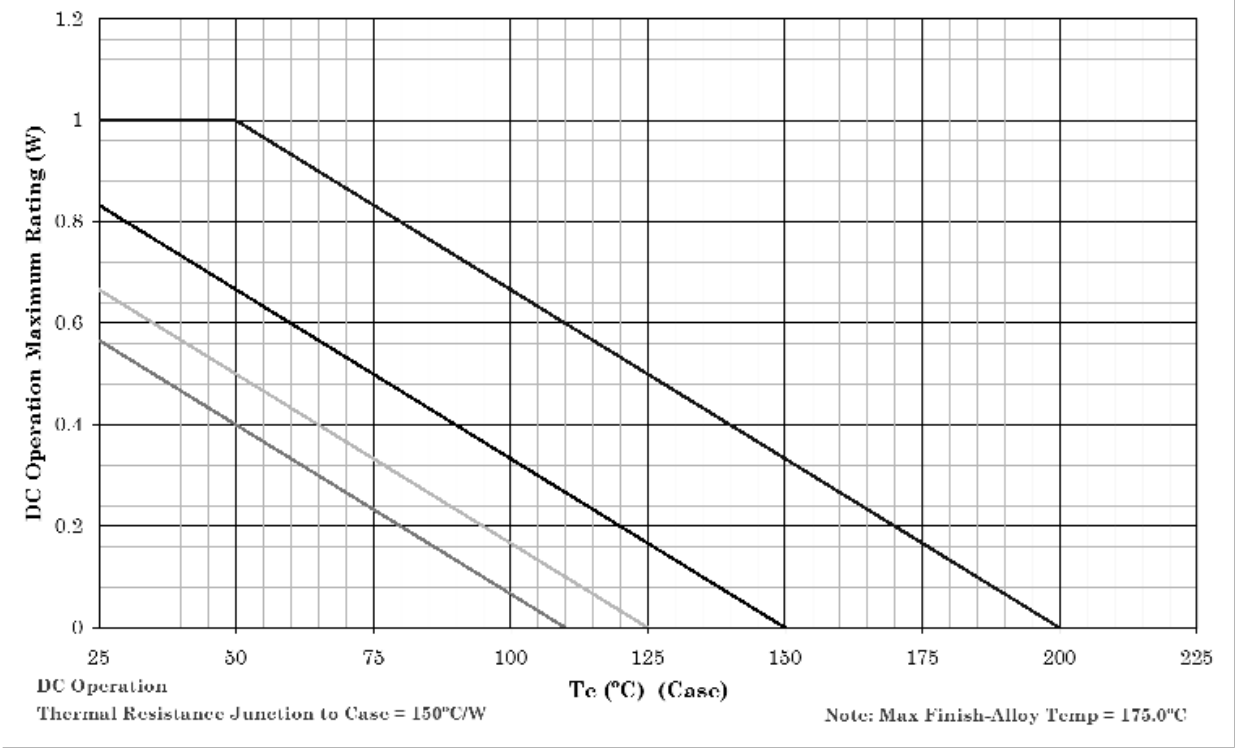
NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperatures and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at T_J ≤ +150°C, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at T_J ≤ +125°C, and +110°C to show power rating where most users want to limit T_J in their application.

FIGURE 12. Derating for 2N3057A ($R_{\theta JC}$) (TO-46), base case mounted.

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TEMPERATURE-POWER DERATING CURVE
T_C = 25°C; DEVICE TYPE 2N3700



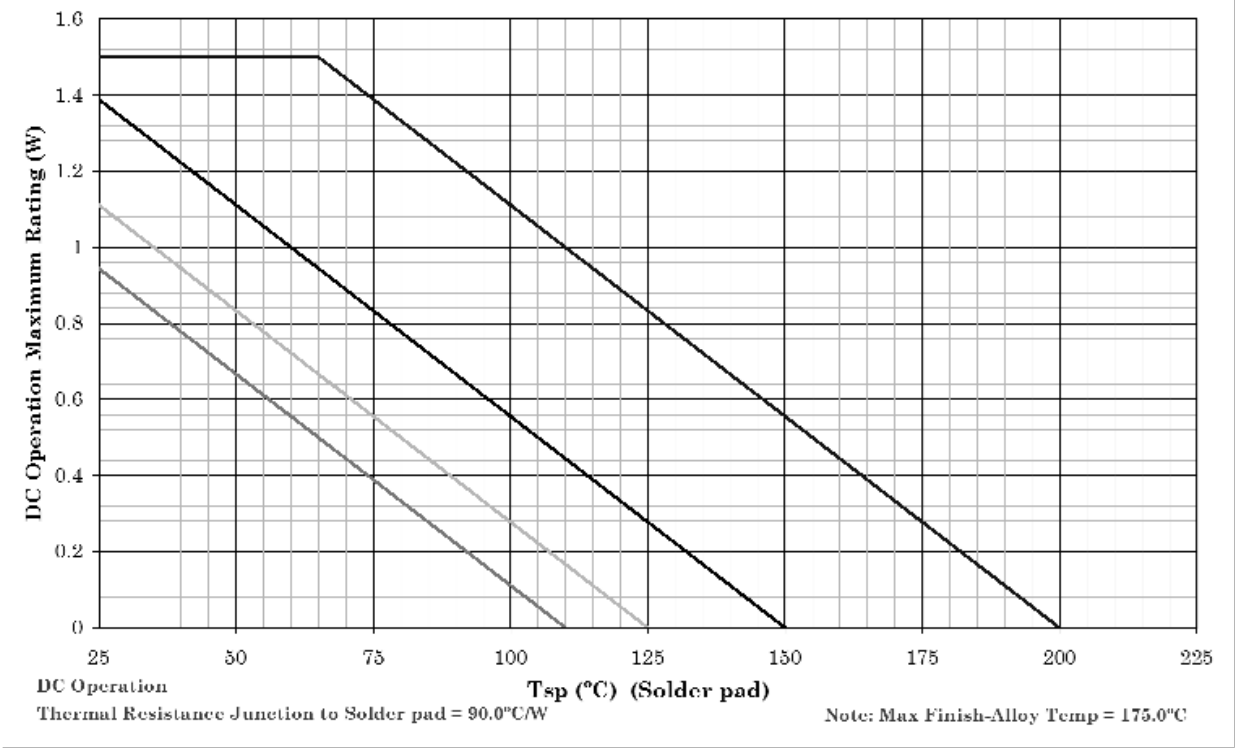
NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperatures and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at T_J ≤ +150°C, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at T_J ≤ +125°C, and +110°C to show power rating where most users want to limit T_J in their application.

FIGURE 13. Derating for 2N3700 ($R_{\theta JC}$) (TO-18), base case mounted.

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TEMPERATURE-POWER DERATING CURVE
T_{SP} = 25°C; DEVICE TYPE 2N3700UB AND 2N3700UBC

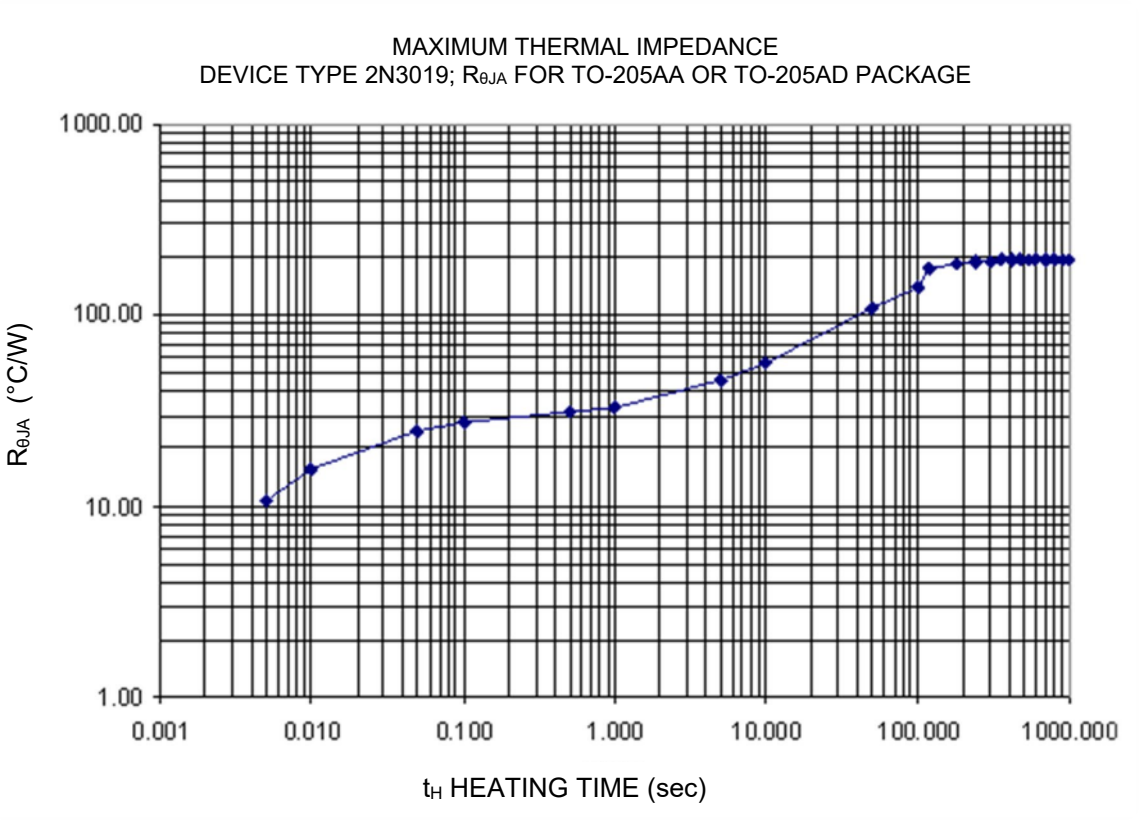


NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperatures and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at T_J ≤ +150°C, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at T_J ≤ +125°C, and +110°C to show power rating where most users want to limit T_J in their application.

FIGURE 14. Derating for 2N3700UB and 2N3700UBC ($R_{\theta JSP(IS)}$), infinite sink 3-points.

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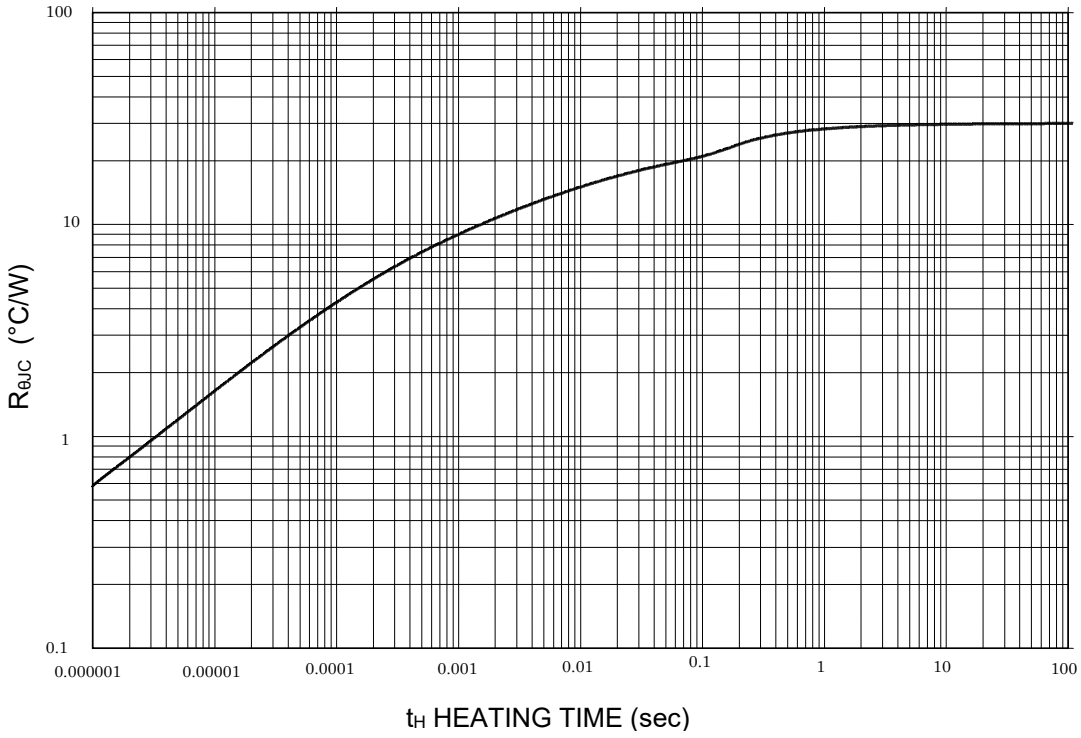


$R_{\theta JA}$ = 195 °C/W

FIGURE 15. Thermal impedance graph ($R_{\theta JA}$) for a 2N3019 in a TO-205AA or TO-205AD package.

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MAXIMUM THERMAL IMPEDANCE
DEVICE TYPE 2N3019;
$R_{\theta JC}$ FOR TO-205AA OR TO-205AD PACKAGE

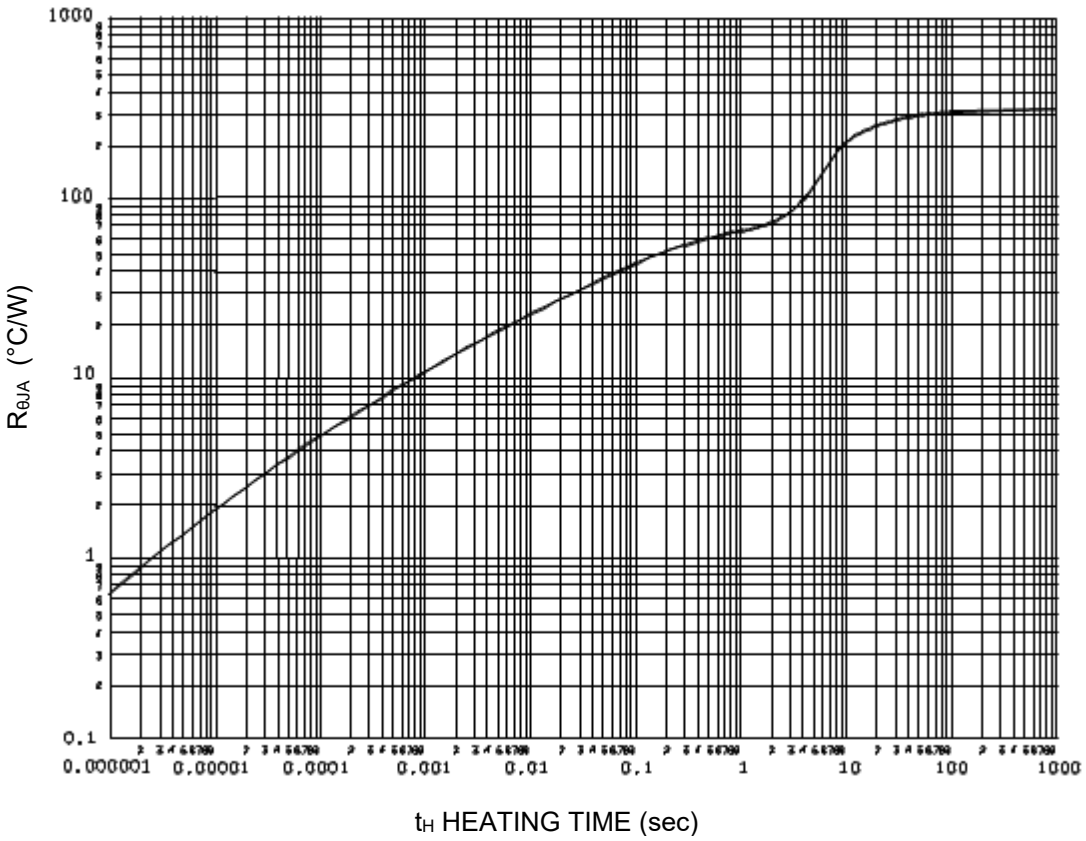


$R_{\theta JC}$ = 30 °C/W

FIGURE 16. Thermal impedance graph ($R_{\theta JC}$) for a 2N3019 in a TO-205AA or TO-205AD package.

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Check the source to verify that this is the current version before use.

MAXIMUM THERMAL IMPEDANCE
DEVICE TYPE 2N3057A;
$R_{\theta JA}$ FOR TO-206AB PACKAGE

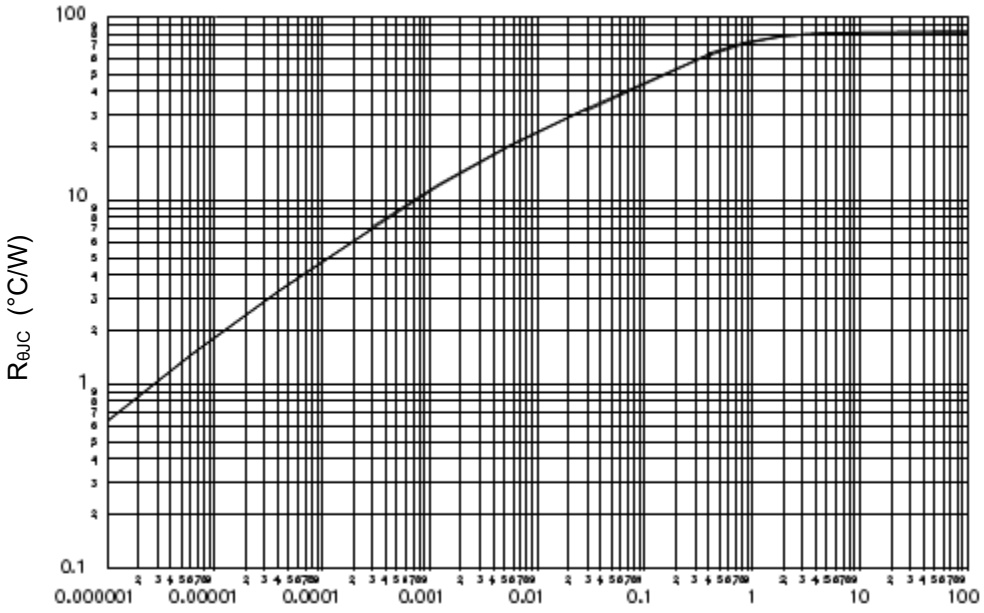


$R_{\theta JA}$ = 325 °C/W

FIGURE 17. Thermal impedance graph ($R_{\theta JA}$) for devices (2N3057A) in TO-206AB package.

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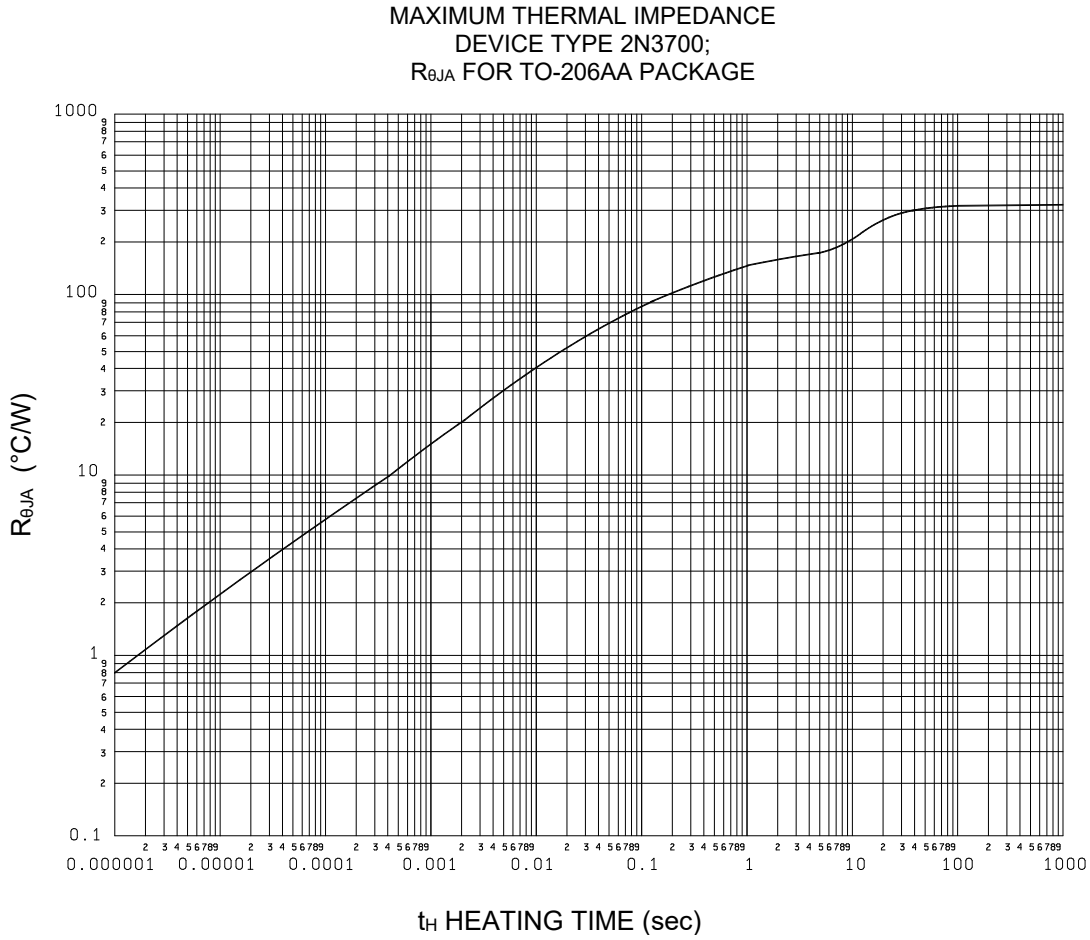
MAXIMUM THERMAL IMPEDANCE
DEVICE TYPE 2N3057A;
$R_{\theta JC}$ FOR TO-206AB PACKAGE



$R_{\theta JC}$ = 80 °C/W

FIGURE 18. Thermal impedance graph ($R_{\theta JC}$) for devices (2N3057A) in TO-206AB package.

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Check the source to verify that this is the current version before use.

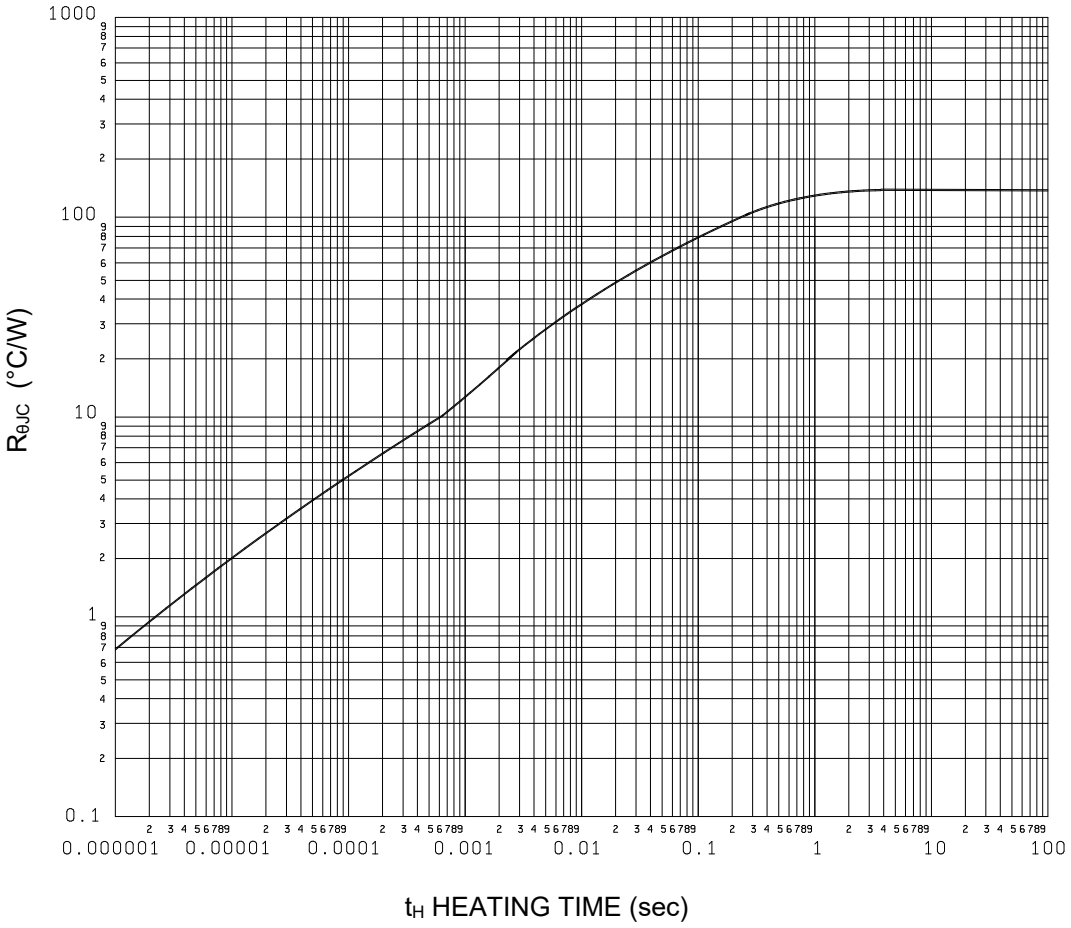


$R_{\theta JA}$ = 325 °C/W

FIGURE 19. Thermal impedance graph ($R_{\theta JA}$) for devices (2N3700) in a TO-206AA package.

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Check the source to verify that this is the current version before use.

Maximum Thermal Impedance
DEVICE TYPE 2N3700;
$R_{\theta JC}$ FOR TO-206AA PACKAGE

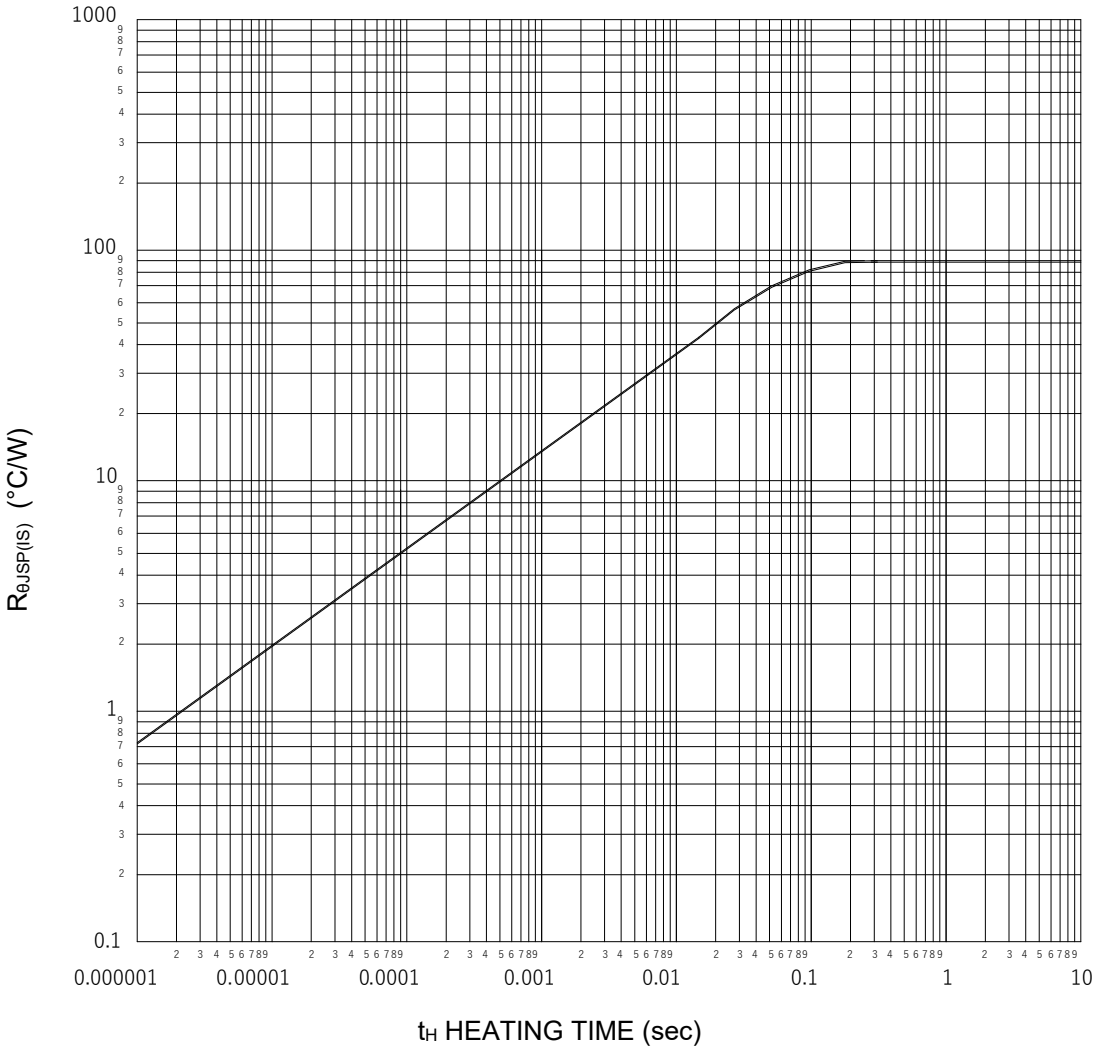


$R_{\theta JC}$ = 150 °C/W

FIGURE 20. Thermal impedance graph ($R_{\theta JC}$) for devices (2N3700) in a TO-206AA package.

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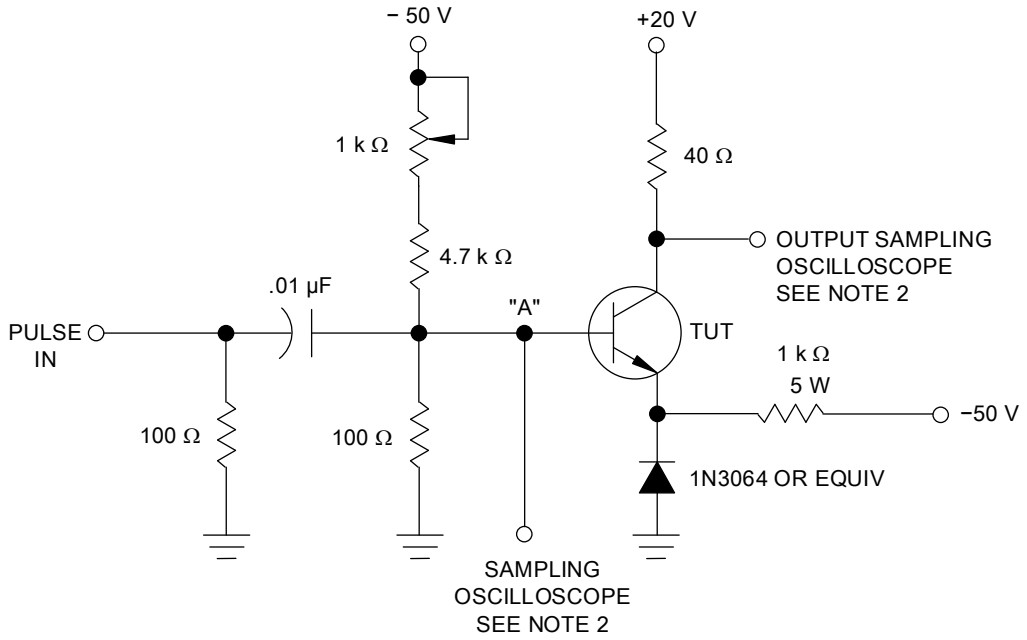
Maximum Thermal Impedance
DEVICE TYPE 2N3700;
$R_{\theta JSP\ (IS)}$ FOR UB and UBC PACKAGE

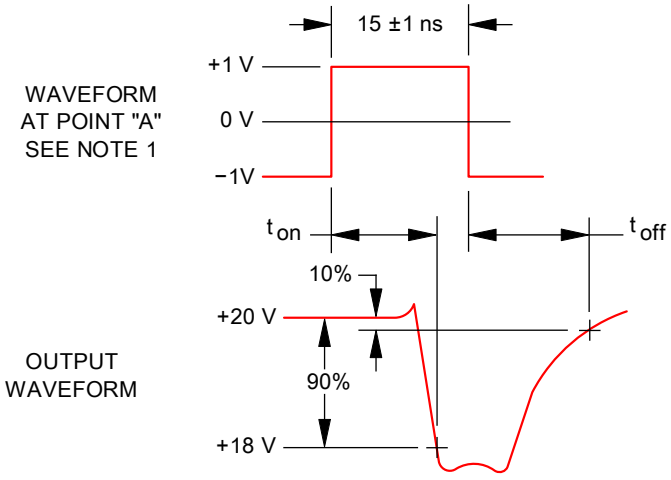


$R_{\theta JSP(IS)}$ = 90 °C/W

FIGURE 21. Thermal impedance graph ($R_{\theta JSP(IS)}$) for a 2N3700 in a UB and UBC package.

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Check the source to verify that this is the current version before use.





NOTES:

1. The rise time (t_r) of the applied pulse shall be ≤ 2.0 ns, duty cycle ≤ 2 percent and the generator source impedance shall be 50 ohms.
2. Sampling oscilloscope: Z_{IN} ≥ 100 kΩ, C_{IN} ≤ 12 pF, rise time ≤ 2.0 ns.

FIGURE 22. Non-saturated switching-time test circuit.

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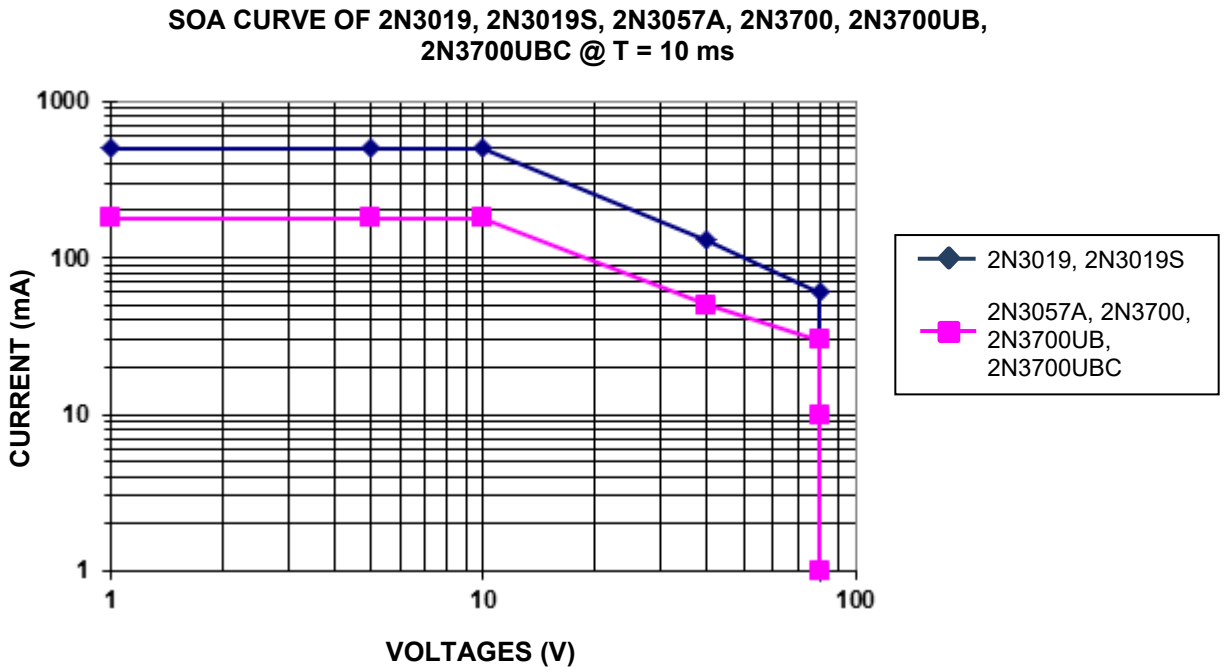


FIGURE 23. Maximum safe operating graph (10 ms).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

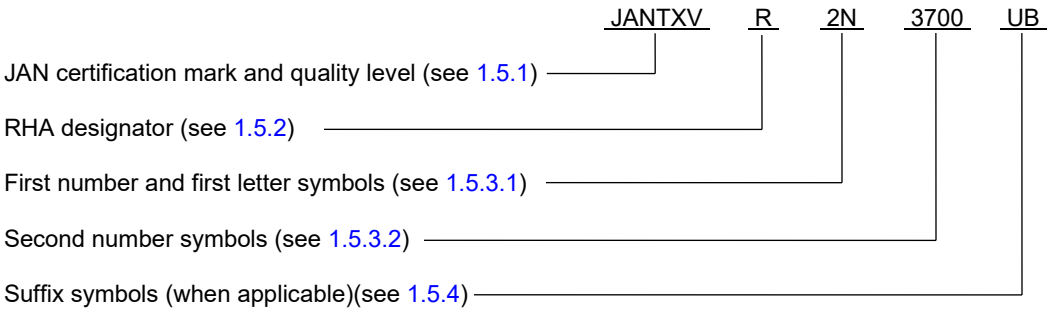
d. The complete PIN, see 1.5 and 6.4.

e. For acquisition of RHA designed devices, table II, subgroup 1 testing of group D is optional. If table II, subgroup 1 testing is desired, it must be specified in the contract.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

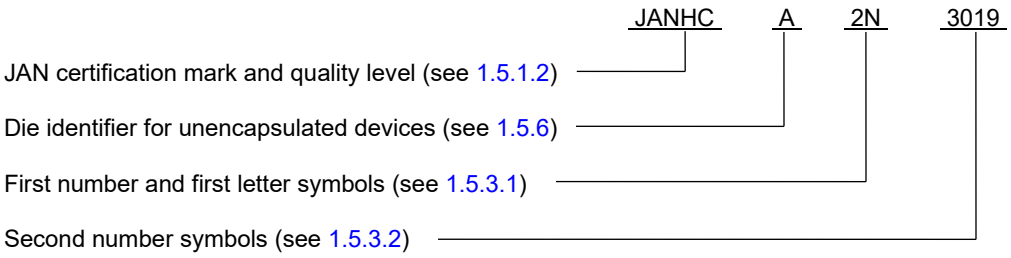
6.4 PIN construction examples.

6.4.1 Encapsulated devices The PINs for encapsulated devices are constructed using the following form.



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6.4.2 Un-encapsulated devices. The PINs for un-encapsulated devices are constructed using the following form.



6.5 List of PINs.

6.5.1 PINs for encapsulated devices. The following is a list of possible PINs for encapsulated devices available on this specification sheet.

PINs for types 2N3019 (1)	PINs for types 2N3019S (1)	PINs for types 2N3057A (1)	PINs for types 2N3700 (1)
JAN2N3019	JAN2N3019S	JAN2N3057A	JAN2N3700
JANTX2N3019	JANTX2N3019S	JANTX2N3057A	JANTX2N3700
JANTXV2N3019	JANTXV2N3019S	JANTXV2N3057A	JANTXV2N3700
JANS2N3019	JANS2N3019S	JANS2N3057A	JANS2N3700
JANTXVR2N3019	JANTXVR2N3019S	JANTXVR2N3057A	JANTXVR2N3700
JANTXVF2N3019	JANTXVF2N3019S	JANTXVF2N3057A	JANTXVF2N3700
JANS#2N3019	JANS#2N3019S	JANS#2N3057A	JANS#2N3700

(1) The number sign (#) represents one of eleven RHA designators available (E, K, U, M, D, P, L, R, F, G, or H).

PINs for types 2N3700UB (1)	PINs for types 2N3700UBC (1)
JAN2N3700UB	JAN2N3700UBC
JANTX2N3700UB	JANTX2N3700UBC
JANTXV2N3700UB	JANTXV2N3700UBC
JANS2N3700UB	JANS2N3700UBC
JANTXVR2N3700UB	JANTXVR2N3700UBC
JANTXVF2N3700UB	JANTXVF2N3700UBC
JANS#2N3700UB	JANS#2N3700UBC

(1) The number sign (#) represents one of eleven RHA designators available (E, K, U, M, D, P, L, R, F, G, or H).

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6.5.2 PINs for unencapsulated die. The following is a list of possible PINs for unencapsulated die available on this specification sheet.

PINs for types 2N3019 (1)		PINs for types 2N3700 (1)			
JANHCA2N3019	JANHCD2N3019	JANHCA2N3700	JANHCB2N3700	JANHCC2N3700	JANHCD2N3700
JANKCA2N3019	JANKCD2N3019	JANKCA2N3700	JANKCB2N3700	JANKCC2N3700	JANKCD2N3700
JANHCA#2N3019	JANHCD#2N3019	JANHCA#2N3700	JANHCB#2N3700	JANHCC#2N3700	JANHCD#2N3700
JANKCA#2N3019	JANKCD#2N3019	JANKCA#2N3700	JANKCB#2N3700	JANKCC#2N3700	JANKCD#2N3700

(1) The number sign (#) represents one of eleven RHA designators available (E, K, U, M, D, P, L, R, F, G, or H).

6.6 Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example JANHCA2N3700) will be identified on the QML.

Die ordering information				
PIN	Manufacturer			
	34156	43611	52GC4	52GC4
2N3700	JANHCA2N3700 JANKCA2N3700	JANHCB2N3700 JANKCB2N3700	JANHCC2N3700 JANKCC2N3700	JANHCD2N3700 JANKCD2N3700
2N3019	JANHCA2N3019 JANKCA2N3019			JANHCD2N3019 JANKCD2N3019

6.7 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at Semiconductor@dla.mil or by facsimile (614) 693-1642 or DSN 850-6939.

6.8 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Army − CR
Navy − SH
Air Force − 85
NASA − NA
DLA − CC

Review activities:
Army − AR, MI, SM
Navy − AS, MC
Air Force − 19

Preparing activity:
DLA − CC

(Project 5961-2023-064)

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